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Original Article

Longitudinal impact of a pharmacist-led stepwise antimicrobial stewardship program on intravenous antimicrobial use at a small resource-limited Japanese hospital

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ABSTRACT

Background: Clinical pharmacists play a central role in the implementation of antimicrobial stewardship programs (ASP). However, personnel shortages, the lack of a system for career development, and other challenges impede the active contribution of clinical pharmacists to ASPs in Japan.
Aim: To conduct a long-term evaluation of a pharmacist-led antimicrobial stewardship program.
Methods: The present, retrospective study used segmented time-series analysis to examine antimicrobial consumption, blood culture practices, and the rate of resistance in pathogens to the antimicrobials prescribed.
Findings: After the ASP was introduced, days of therapy (DOT) with anti-pseudomonal agents demonstrated a downward trend (−0.31; 95 % confidence interval [CI]: −0.43 to −0.19; $p < 0.01$). Moreover, the trend in the number of blood cultures after the intervention significantly increased (0.20; 95 % CI: 0.05 to 0.35; $p = 0.01$), and the rate of resistance of *Pseudomonas aeruginosa* to cefepime and piperacillin decreased. No significant change was observed in the DOT trend for total intravenous antimicrobial use before or after the intervention (−1.07; 95 % CI: −2.44 to 0.29; $p = 0.12$).
Conclusion: Although the role of pharmacists in an ASP should continue to be examined, the present, longitudinal study found that stepwise introduction of an ASP reduced broad-spectrum antimicrobial use and was associated with a decrease in the antimicrobial resistance rate of *Pseudomonas aeruginosa* but not of third-generation cephalosporin-resistant *Escherichia coli* or MRSA.

1. Introduction

Antimicrobial resistance (AMR) is a significant public health issue. Many nations have developed an AMR action plan incorporating various strategies to address this challenge [1]. The Japanese AMR action plan was launched in 2016 [2], and acute care hospitals are now required to implement an antimicrobial stewardship program (ASP) [3,4]. Clinical pharmacists play a central role in ASPs by collaborating with infectious disease specialists [5]. However, owing to the shortage of infectious disease specialists in Japan, non-infectious disease specialists across various healthcare settings are made to serve as the lead physicians. However, the latter often lack expertise in managing infectious diseases and are unable to dedicate sufficient time to ASP activities, thus forcing pharmacists to take the lead in these programs [6–8].

While there are numerous studies of the short-term effects of pharmacist-led ASPs on antimicrobial use [9–11], the data on the long-term effects are scarce, particularly in small to medium-sized hospitals in Japan, which make up a significant proportion of the nation's healthcare institutions [3,12]. Furthermore, barriers, such as human resource shortages and the lack of career development, have been identified as obstacles to the participation of clinical pharmacists in ASPs, especially in Asian countries, including Japan [5].

Given these challenges, the present study implemented a stepwise pharmacist-led ASP and retrospectively evaluated its long-term effects in a small, acute-care hospital in Japan. The evaluation focused on antimicrobial use, the rate of antimicrobial resistance, and blood culture

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practices with the aim of providing insights into the effectiveness of pharmacist-led ASPs in resource-limited settings.

2. Materials and methods

2.1. Study setting

The present study was conducted at Tama-Nambu Chiiki Hospital, a community hospital with 287 beds and no infectious disease specialist. To assess the impact of the ASP, segmented time-series analysis was used to compare a 92-month period following the introduction of an ASP (August 2015–March 2023) with a 28-month, pre-ASP period (April 2013–July 2015).

2.2. ASP

The ASP was introduced in stages starting in August 2015 under the leadership of a pharmacist. Prior to its implementation, the pharmacist discussed the content of the ASP with physicians, laboratory technicians, and nurses, who later formed a team for implementing the program with the following fulltime equivalents (FTE): pharmacist, 0.2–1.0; physicians, 0.05; laboratory technicians, 0.1; and nurses, 0.05. In April 2020 and November 2021, respectively, a locum infectious disease physician (0.05 FTE) and an infectious disease specialist (0.05 FTE) providing consultations joined the ASP. The key components of the ASP included prospective audit and feedback (PAF), post-prescription review and feedback (PPRF), pre-authorization of specific antimicrobials, treatment management interventions for patients with a positive blood culture (syndrome-specific intervention), the introduction of hospital-specific guidelines, and a reappraisal of the antimicrobials used in the hospital (Table 1).

2.2.1. PAF and PPRF

Initially, the PAF targeted intravenous carbapenems. Later, a transition was made from this method to PPRF, which was expanded to cover all intravenous antimicrobials. In both the PAF and PPRF, the pharmacist reviewed patients who had received a prescription of a targeted antimicrobial on an audit day using electronic medical records and provided feedback to the attending or prescribing physician if an antimicrobial was deemed unnecessary or inappropriate for the diagnosis, culture results or hospital-specific guidelines.

Table 1
Details of the implementation of the antimicrobial stewardship program.

Date	Details of intervention
August 2015	Daily, pharmacist-driven prospective audit and feedback for carbapenems Reducing redundancies or inefficiencies in 4th generation cephalosporin use
September 2018	Discontinuation of sulbactam-cefoperazone use at the study center
April 2019	Daily, pharmacist-driven post-prescription review and feedback for all intravenous antimicrobials at 72 h
May 2019	Preauthorization of ceftazidime use by pharmacist
April 2020	Weekly, pharmacist-driven post-prescription review and feedback for all intravenous antimicrobials Preauthorization of fosfomycin use by pharmacist
September 2020	Implementation of local, institutional guidelines
August 2021	Weekly post-prescription review and feedback on anti-pseudomonas agent use with an infectious disease physician and pharmacist
April 2022	Preauthorization of levofloxacin Introduction of aztreonam use
September 2022	Implementation of surveys on compliance with institutional guidelines and feedback

2.2.2. Pre-authorization of antimicrobials

The pre-authorization policy applied to intravenous fluoroquinolones, specifically levofloxacin.

2.2.3. Intervention in patients with a positive blood culture result

Interventions in patients with a positive blood culture result included assessment for any discrepancy between the prescribed antimicrobial(s) and the culture result, recommending adjustments whenever necessary, and confirming negative conversion. Blood cultures were collected by either the physicians or nurses, with the nurses being the primary practitioners. No specific measures were taken to change their role in blood culture collection during the study period.

In the PPRF, pre-authorization, and blood culture intervention processes, pharmacists were primarily responsible for reviewing the electronic medical records, documenting their findings, and contacting the attending or prescribing physician by telephone.

2.2.4. Hospital-specific guidelines

Hospital-specific guidelines were formulated in September 2020 to handle infections that are frequently encountered in the hospital setting. The guidelines were made accessible through the electronic medical record system. In September 2022, a pocket guide for inpatient care was created and distributed to all clinicians. Additionally, quarterly feedback on compliance with the guidelines during the initial administration of intravenous antimicrobials and initial testing was provided directly to each department. Compliance surveys and assessments were primarily conducted by the pharmacist, with the infectious disease specialists making treatment decisions in cases of infection not covered by the guidelines.

2.2.5. Review of antimicrobial agents approved for use at the study center

As part of the review of the antimicrobial agents approved for use, multiple, fourth-generation cephalosporins were streamlined into a single option.

2.3. Data collection

The present study was based on data from the study center and the Japan Surveillance for Infection Prevention and Healthcare Epidemiology (J-SIPHE) [13], including information obtained from records of medical treatments and procedures linked to billing data under the Diagnosis Procedure Combination system [14]. Antimicrobial consumption patterns were able to be analysed by calculating DOT using the data from J-SIPHE.

Days of antibiotic spectrum coverage (DASC) were calculated using the ASC scores reported in previous studies [15,16]. DASC/DOT, which represents the average antimicrobial spectrum provided to inpatients, was also calculated. Patients were excluded if they were receiving an antiviral or antifungal agent for which the ASC scores had yet to be determined. The ASC score was defined in accordance with the method described in the previously cited studies [15] by evaluating *in vitro* antimicrobial activity against naive organisms and the potential of these organisms to develop resistance on exposure. Table 2 shows the detailed definition of the scores for the target antimicrobials.

The resistance rate was defined as the percentage of isolates with non-susceptibility (intermediate or resistant). The detection of the same species within one month in the same patient was excluded from analysis to avoid duplication. Identification and sensitivity testing were performed using MicroScan WalkAway-40 Plus (Siemens, Germany). Susceptibility was interpreted in accordance with document M100-S27 of the Clinical and Laboratory Standards Institute.

2.4. Outcomes

The process and outcome metrics were treated as the study outcomes. The process metrics included any change in the monthly days of

Table 2
Antimicrobial spectrum coverage score in the present study.

Antimicrobial class	Antimicrobials	ASC score[a]	*S. aureus*	*Strepto-coccus* spp.	*E. faecalis*	Anaer-obes (Oral)	*B. fragilis*	*Moraxella* / *H. influenzae*	*E. coli K. pneumoniae*	*Enterobacter* / *Serratia* / *Citrobacter*	*P. aeruginosa*	*A. baumannii*	Atypical	ESBL	MRSA	PRSP	VRE	CRE
Cephalosporin	Cefpirome	8	1	1	0	0	0	1	1	1	1	1	0	0	0	1	0	0
Aminoglycoside	Arbekacin	9	1	0	0	0	0	1	1	1	1	1	0	1	1	0	0	1
Aminoglycoside	Kanamycin	9	1	0	0	0	0	1	1	1	1	1	0	1	1	0	0	1
Fluoroquinolone	Pazufloxacin	9	1	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0

Dichotomous classifications were assigned to each antibiotic-organism combination, and the ASC scores were determined following the format presented in Table 2 of the original study by Kakiuchi et al. [PMID: 34910130].

Abbreviations: ASC score, antimicrobial spectrum coverage score; *S. aureus*, *Staphylococcus aureus*; *E. faecalis*, *Enterococcus faecalis*; *B. fragilis*, *Bacteroides fragilis*; *H. influenzae*, *Haemophilus influenzae*; *E. coli*, *Escherichia coli*; *K. pneumoniae*, *Klebsiella pneumoniae*; *P. aeruginosa*, *Pseudomonas aeruginosa*; *A. baumanii*, *Acinetobacter baumanii*; ESBL, extended-spectrum β-lactamase-producing Enterobacterales; MRSA, methicillin-resistant *S. aureus*; PRSP, penicillin-resistant *Streptococcus pneumoniae*; VRE, vancomycin-resistant *Enterococcus faecium*; CRE, carbapenem-resistant Enterobacterales.

[a] The ASC score is the sum of all the points in each row.

therapy (DOT) per 1000 patients-days (PD) for all intravenous antimicrobial agents, anti-methicillin-resistant *Staphylococcus aureus* (MRSA) agents (vancomycin, teicoplanin, linezolid, daptomycin, and arbekacin), anti-pseudomonal agents (carbapenems, piperacillin-tazobactam, fourth generation cephalosporins, cefoperazone-sulbactam), and quinolones in the DASC, DASC/DOT for all intravenous antimicrobial agents, and the number of blood culture sets per 1000 PD between the pre-intervention and the post-intervention periods. The outcome measures were the monthly incidence of MRSA and third-generation, cephalosporin-resistant *E. coli* isolated from clinical samples, the rate of resistance of *Pseudomonas aeruginosa* to piperacillin, cefepime, aztreonam, meropenem, imipenem-cilastatin, gentamicin, and amikacin; the average length of stay; and monthly in-hospital mortality. When evaluating the rate of antimicrobial resistance of *Pseudomonas aeruginosa*, the rate of resistance to each antimicrobial agent was used as an outcome measure instead of the incidence of resistant bacteria because the number of bacteria resistant to two or more antimicrobial agents was too small to constitute a meaningful outcome. In-hospital mortality was defined as deaths occurring among all hospitalized patients, regardless of infection status or antimicrobial treatment.

2.5. Statistical analysis

Segmented regression in interrupted time-series analysis (ITSA) [17] was used to assess changes in the monthly DOT per 1000 PD, DASC per 1000 PD, DASC/DOT, number of blood culture sets per 1000 PD, monthly incidence of MRSA and third generation cephalosporin-resistant *E. coli*, average length of stay, and monthly in-hospital mortality. The ITSA value based on monthly intervals was 28 and 92 before and after the implementation of the ASP, respectively. The Cumby-Huizinga test for autocorrelation was performed, and the resistance rate per year was analysed using the Cochran-Armitage test for trends. $P < 0.05$ was considered to indicate statistical significance. Stata version 17.0 (StataCorp, College Station, TX) was used for all statistical analyses. The requirement for patient consent was waived because the study was an institutional quality-improvement project. The institutional review board at Tokyo Metropolitan Tama-Nambu Chiiki Hospital approved this study.

3. Results

Fig. 1 and Table 3 show the results of the ITSA of the changes in DOT for all as well as individual antimicrobial agents. DOT with anti-pseudomonal agents demonstrated a downward trend after the introduction of the ASP (−0.31; 95 % confidence interval [CI]: −0.43 to −0.19; $p < 0.01$). In contrast, DOT for all intravenous antimicrobials demonstrated no significant change before or after the intervention (−1.07; 95 % CI: −2.44 to 0.29; $p = 0.12$). The trend in DOT with anti-MRSA agents showed a statistically significant increase before and after the introduction of the ASP (0.13; 95 % CI: 0.05 to 0.20; $p < 0.01$).

Where antipseudomonal agents were concerned, DOT with carbapenems demonstrated a declining trend prior to the intervention (−0.16; 95 % CI: −0.31 to −0.02; $p = 0.03$) but remained stable after ASP implementation (0.01; 95 % CI: −0.01 to 0.04; $p = 0.35$), with the post-intervention trend indicating a statistically significant increase (0.18; 95 % CI: 0.03 to 0.32; $p = 0.02$). Conversely, the trend in piperacillin-tazobactam use increased before the intervention (0.56; 95 % CI: 0.27 to 0.84; $p < 0.01$) but stabilized after the intervention (−0.06; 95 % CI: −0.17 to 0.04; $p = 0.21$) before significantly decreasing (−0.62; 95 % CI: −0.93 to −0.32; $p < 0.01$). The trend in DASC per 1000 PD and DASC/DOT significantly decreased post-intervention (−3.46; 95 % CI: −5.33 to −1.59; $p < 0.01$ and −0.01; 95 % CI: −0.02 to 0.01; $p < 0.01$, respectively).

Table 4 shows changes in the outcome indicators and number of blood culture sets per 1000 PD. The trend in the average length of hospital stay decreased before the intervention (−0.09; 95 % CI: −0.14

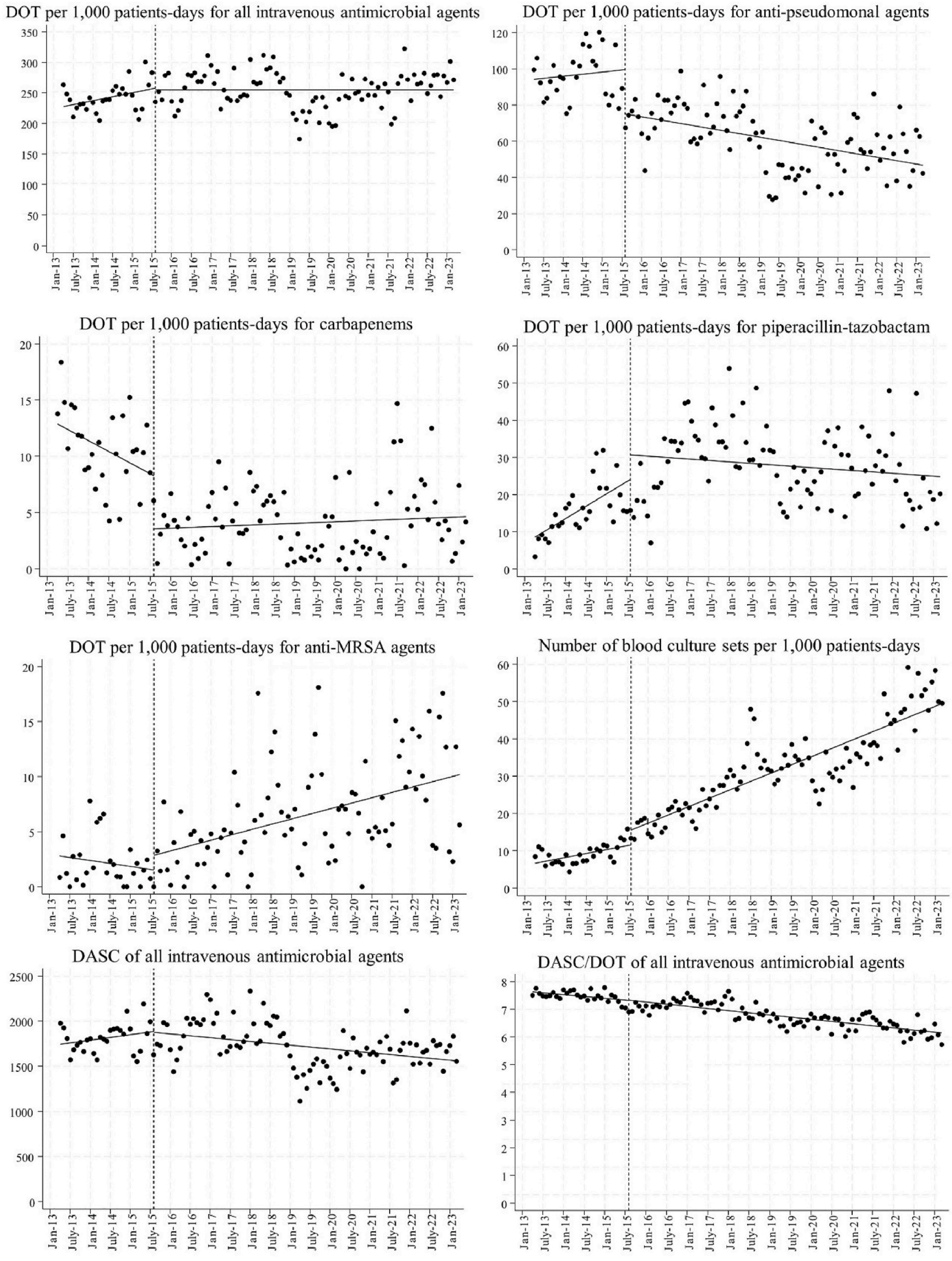

Fig. 1. Changes in antimicrobial use and the number of blood culture sets before and after antimicrobial stewardship program implementation.

to −0.05; $p < 0.01$) and slowed after the intervention (−0.01; 95 % CI: −0.01 to 0.01; p = 0.07). The trend in the mortality rate increased prior to the intervention (0.04; 95 % CI: 0.01 to 0.07; p = 0.03) but significantly improved post-intervention (−0.04; 95 % CI: −0.07 to −0.01; p = 0.04). The rate of detection of MRSA and third-generation cephalosporin-resistant *E. coli* demonstrated an increasing trend. Moreover, the trend in the number of blood culture tests performed after the intervention significantly increased (0.20; 95 % CI: 0.05 to 0.35; p = 0.01).

Table 3
Changes in antimicrobial use before and after the implementation of the antimicrobial stewardship program analysed using interrupted time series analysis.

Antimicrobials	Baseline trend in preintervention period (95 % CI)	*p*	Slope in intervention period (95 % CI)	*p*	Change in slope	*p*
Monthly DOT per 1000 PD						
Antipseudomonal agents	0.19 (−0.46 to 0.84)	0.57	−0.31 (−0.43 to −0.19)	**<0.01**	−0.50 (−1.16 to 0.16)	0.14
Penicillins	0.11 (−0.04 to 0.26)	0.14	0.01 (−0.05 to 0.05)	1.0	−0.11 (−0.27 to 0.04)	0.16
Ampicillin/sulbactam	−0.05 (−0.49 to 0.38)	0.81	−0.24 (−0.34 to −0.15)	**<0.01**	−0.19 (−0.63 to 0.25)	0.40
Piperacillin/tazobactam	0.56 (0.27–0.84)	**<0.01**	−0.06 (−0.17 to 0.04)	0.21	−0.62 (−0.93 to −0.32)	**<0.01**
G1 Cephem	0.47 (0.35–0.60)	**<0.01**	0.27 (0.21–0.33)	**<0.01**	−0.20 (−0.34 to −0.06)	**<0.01**
G2 Cephem	0.17 (0.05–0.30)	**<0.01**	−0.03 (−0.03 to −0.02)	**<0.01**	−0.20 (−0.33 to −0.07)	**<0.01**
Cephamycin	−0.17 (−0.55 to 0.21)	0.37	0.08 (0.02–0.15)	**<0.01**	0.25 (−0.13 to 0.64)	0.19
G3 Cephem	0.09 (−0.15 to 0.34)	0.46	0.50 (0.38–0.62)	**<0.01**	0.41 (0.13–0.68)	**<0.01**
G4 Cephem	0.28 (−0.20 to 0.75)	0.25	−0.07 (−0.16 to 0.02)	0.11	−0.35 (−0.83 to 0.13)	0.16
Carbapenems	−0.16 (−0.31 to −0.02)	**0.03**	0.01 (−0.01 to 0.04)	0.35	0.18 (0.03–0.32)	**0.02**
Quinolones	−0.06 (−0.16 to 0.03)	0.19	−0.04 (−0.06 to −0.01)	**<0.01**	0.03 (−0.07 to 0.12)	0.61
Anti-MRSA	−0.05 (−0.11 to 0.02)	0.19	0.08 (0.05–0.11)	**<0.01**	0.13 (0.05–0.20)	**<0.01**
Aminoglycoside	−0.14 (−0.19 to −0.08)	**<0.01**	−0.01 (−0.02 to 0.01)	0.85	0.13 (0.07–0.19)	**<0.01**
All antimicrobials	1.07 (−0.28 to 2.42)	0.12	0.01 (−0.24 to 0.24)	1.0	−1.07 (−2.44 to 0.29)	0.12
Metrics regarding DASC						
DASC per 1000 PD	5.08 (−4.16 to 14.33)	0.28	−3.46 (−5.33 to −1.59)	**<0.01**	−8.54 (−17.95 to 0.86)	0.08
DASC per DOT	−0.01 (−0.02 to −0.01)	**0.01**	−0.01 (−0.02 to 0.01)	**<0.01**	−0.01 (−0.01 to 0.01)	0.66

Abbreviations: PD, patient-Days; CI, confidence interval; DOT, days of therapy; MRSA, methicillin-resistant *Staphylococcus aureus*; DASC, days of antimicrobial spectrum coverage; G, generation.

Table 4
Change in the outcomes, incidence of selected multidrug-resistant organisms, and blood culture sets after antimicrobial stewardship program implementation.

Outcomes	Baseline trend in preintervention period (95 % CI)	*p*	Slope in intervention period (95 % CI)	*p*	Change in slope	*p*
Average length of hospital stay	−0.09 (−0.14 to −0.05)	**<0.01**	−0.01 (−0.01 to 0.01)	0.07	0.09 (0.04–0.13)	**<0.01**
Average monthly mortality rate	0.04 (0.01–0.07)	**0.03**	0.01 (−0.01 to 0.01)	0.57	−0.04 (−0.07 to −0.01)	**0.04**
MRSA incidence per 1000 PD	−0.01 (−0.02 to 0.01)	0.14	0.01 (−0.01 to 0.01)	0.11	0.01 (−0.01 to 0.02)	0.05
Incidence of G3 cephem-resistant *E. coli* per 1000 PD	−0.01 (−0.02 to 0.01)	0.24	0.01 (0.01–0.01)	**<0.01**	0.01 (0.01–0.03)	**0.03**
Blood culture sets per 1000 PD	0.18 (0.03–0.32)	**0.02**	0.37 (0.33–0.42)	**<0.01**	0.20 (0.05–0.35)	**0.01**

Abbreviations: CI, confidence interval; PD, patient-days; MRSA, methicillin-resistant *Staphylococcus aureus*; G, generation.

Table 5
Rate of resistance of *P. aeruginosa* to antimicrobials in 2013–2022 (%).

Characteristics	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	*p*
Piperacillin	1.4 (1/74)	16.7 (3/18)	12.0 (13/108)	14.9 (15/101)	3.7 (4/107)	6.9 (9/130)	7.4 (11/148)	5.4 (6/112)	8.1 (10/123)	1.5 (2/134)	**0.03**
Cefepime	8.1 (6/74)	22.2 (4/18)	8.3 (9/108)	11.0 (11/100)	1.9 (2/107)	3.8 (5/130)	5.4 (8/148)	7.1 (8/112)	6.5 (8/123)	3.0 (4/134)	**0.01**
Aztreonam	12.2 (9/74)	22.2 (4/18)	17.6 (19/108)	21.8 (22/101)	13.1 (14/107)	9.2 (12/130)	11.5 (17/148)	14.3 (16/112)	10.6 (13/123)	12.7 (8/63)	0.05
Meropenem	4.1 (3/74)	11.1 (2/18)	5.6 (6/107)	5.1 (5/99)	0.9 (1/106)	3.1 (4/130)	2.0 (3/148)	3.6 (4/112)	5.7 (7/123)	1.4 (2/145)	0.10
Imipenem/cilastatin	4.1 (3/74)	5.6 (1/18)	12.3 (13/106)	5.9 (6/101)	4.7 (5/107)	4.7 (6/129)	5.4 (8/148)	6.3 (7/112)	8.9 (11/123)	9.4 (3/32)	0.39
Gentamicin	10.8 (8/74)	27.8 (5/18)	12.0 (13/108)	15.8 (16/101)	3.7 (84/107)	10.0 (13/130)	12.8 (19/148)	9.8 (11/112)	15.4 (19/123)	9.1 (5/55)	0.44
Amikacin	1.4 (1/74)	5.6 (1/18)	1.9 (2/108)	4.0 (4/101)	1.9 (2/107)	3.1 (4/130)	2.7 (4/148)	0.9 (1/112)	4.1 (5/123)	0.00 (0/92)	0.35

Table 5 and Fig. 2 show that the resistance of *Pseudomonas aeruginosa* to cefepime and piperacillin decreased.

4. Discussion

The present study, which evaluated the long-term effects of a pharmacist-led ASP in a small, acute-care hospital in Japan, found a significant reduction in the use of broad-spectrum antimicrobials and a shift towards the use of narrower-spectrum agents hospital-wide. An improvement in the antimicrobial resistance rate of *Pseudomonas aeruginosa* was also noted. These findings suggested that even in small hospitals without an infectious disease specialist, a pharmacist-led ASP can contribute to controlling AMR.

Previous studies have indicated the efficacy of pharmacist-led ASPs particularly in reducing the use of broad-spectrum antimicrobials, such as carbapenems and quinolones, in hospitals without an infectious disease specialist [18]. However, the duration of the interventions was shorter (two years) than in the present study, which evaluated a range of indicators besides antimicrobial consumption, including the antimicrobial resistance rate, number of blood cultures, length of hospital stay, and mortality rate. In Japan, the use of antipseudomonal agents is reportedly on the rise [19]. However, the implementation of a pharmacist-led ASP in this study helped reduce the use of antipseudomonal agents, corroborating the findings of previous studies

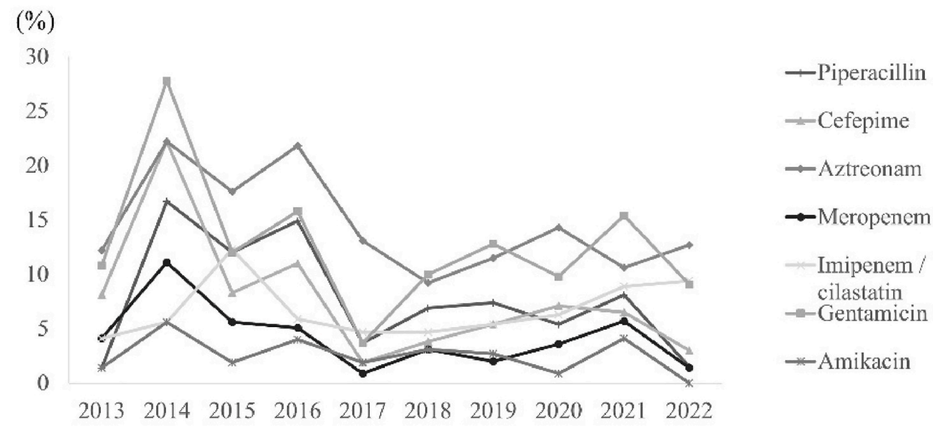

Fig. 2. Rate of resistance of *P. aeruginosa* to antimicrobials in 2013–2022 (%).

which demonstrated the efficacy of a stepwise introduction of ASP in reducing or otherwise improving antimicrobial use [20]. DASC per 1000 PD and DASC/DOT also demonstrated a significantly decreasing trend. Both metrics are recognised as useful indices of ASP efficacy [16,21] and ASP management.

The present study also found that the use of anti-MRSA antimicrobial agents increased significantly after ASP implementation. One possible explanation for this trend is the initiation of proactive interventions in patients with a positive blood culture as part of the ASP. Specifically, this initiative may have encouraged the appropriate use of anti-MRSA antimicrobial agents during empirical therapy and ensured sufficient duration of definitive therapy. A previous study reported that interventions in patients with a positive blood culture were associated with a higher rate of early initiation of anti-MRSA drug therapy and better adherence to appropriate antimicrobial treatment duration [8]. The findings of the present study lend support to the impact of such interventions.

The present study also found an improvement in the rate of resistance of *Pseudomonas aeruginosa* to certain antimicrobials. A previous, seven-year-long, multicentric study also demonstrated a correlation between reduced carbapenem use and an improvement in the rate of resistance of *Pseudomonas aeruginosa* to this class of drug [22], suggesting that long-term intervention is necessary to lessen antimicrobial resistance. However, no reduction in resistance was observed in the incidence of MRSA or third-generation, cephalosporin-resistant *E. coli* possibly because our study included all cases of infectious disease and colonization, including those involving community-acquired strains. Furthermore, addressing the issue of antimicrobial resistance involves not only appropriately dispensing antimicrobial agents but also implementing infection control measures in the hospital and other, related settings [23]. Thus, adherence to infection control measures within the hospital setting need to be analysed in greater detail.

Syndrome-specific interventions in patients with a positive blood culture have been shown to improve the mortality rate [24] and shorten the length of hospital stay [25]. A surveillance study of the blood culture collection rate at 117 acute-care hospitals in Japan found a collection rate of 26.2 sets per 1000 PD, which was far below the 103–188 sets recommended by the American Society for Microbiology [26,27]. Prior to the introduction of the ASP in the present study, the blood culture collection rate was low but increased after the intervention, thanks to timely requests for negative blood culture confirmation by the pharmacists, feedback on adherence to the guidelines, and direct promotion of blood culture collection. Additionally, recommendations for definitive therapy may have led physicians to re-evaluate the need for blood cultures during treatment.

The mortality rate decreased significantly following the intervention despite the continued increase in Japan of the proportion of elderly individuals in the population [28], which generally tends to increase the mortality rate anywhere it occurs [29]. Implementing various ASP activities can contribute to optimizing healthcare for the aging Japanese demographic.

There are several, well-documented barriers to the implementation of pharmacist-led ASPs in Asia, including a shortage of pharmacists who are clinically trained in infectious diseases, the lack of knowledge about ASPs, time constraints, and an insufficient number of dedicated staff [5]. In Japan, training programs for pharmacists in antimicrobial stewardship are limited, and only a small number of institutions have implemented pharmacist-led ASPs [30]. Moreover, the scarcity of infectious disease specialists further restricts opportunities for pharmacists to learn how to implement ASPs effectively in clinical practice. Despite these barriers, our study demonstrated that a stepwise approach to ASP implementation led by a pharmacist resulted in a sustained reduction in the use of broad-spectrum antimicrobials and improvement in the *Pseudomonas aeruginosa* resistance rate. Successful outcomes were achieved by progressively implementing ASP interventions while accumulating experience and refining strategies. This suggests that even in settings with resource constraints, pharmacist-led ASPs are feasible and effective when implemented strategically through a gradual, adaptive approach.

Under these circumstances, direct feedback from pharmacists to physicians not only improves the physicians' knowledge and awareness of the issues but may also contribute to the professional development of the pharmacists themselves. In Japan, where education in infectious diseases and ASP is still in its nascent stage and where the manpower is insufficient to meet the demand, a hands-on approach based on accumulated experience may be viable. Overcoming interdisciplinary barriers through collaboration requires time and resources [31], but most importantly, a proactive, institutional policy on appropriate antimicrobial use is needed.

In this study, the post-prescription audits, feedback, and need for intervention in patients with a positive blood culture were communicated both orally and in writing. A face-to-face approach was adopted in the feedback on adherence to the guidelines. Spoken suggestions were more readily accepted than those submitted in writing [32] and may have contributed to the positive outcomes observed in this study.

5. Limitations

This study has several limitations. First, as it was a monocentric study conducted at a small, acute-care hospital, the results may not be generalizable to larger institutions or other settings. Second, regarding the antimicrobial susceptibility of *Pseudomonas aeruginosa*, the ASP was introduced into a context where the resistance rate was already low. Therefore, it may be difficult to extrapolate the findings to hospitals with a higher, baseline resistance rate. Additionally, the small number of specimens collected in 2014 may have introduced a selection bias. Third, the method of collecting the data on resistance, such as distinguishing between hospital-acquired and community-acquired infections, requires validation. Fourth, although most ASP practices were implemented by a pharmacist, the involvement of an infectious disease specialist following the introduction of the ASP complicated the evaluation of the effect of the pharmacist's role. However, the impact of the infectious disease specialist on the findings may be considered minimal because he was available for only 3 h per week and the pharmacist led the ASP implementation at all other times. Finally, mortality was evaluated using in-hospital mortality rather than mortality specific to patients who received antimicrobial therapy, which may not have been an appropriate outcome measure for assessing the impact of the ASP. However, in-hospital mortality was used to confirm that there was no worsening of patient outcomes following ASP implementation.

6. Conclusion

Despite the nationwide increase in the use of antipseudomonal agents following the implementation of the national AMR action plan, the present study demonstrated that a pharmacist-led ASP in a small, acute-care hospital setting in Japan was effective in continuously reducing DOT with antipseudomonal agents. While an improvement in the rate of resistance of *Pseudomonas aeruginosa* to cefepime and piperacillin was observed, no significant reduction was seen in third-generation cephalosporin-resistant *E. coli* or MRSA. These findings highlight the need for broader, multicentric, pharmacist-led initiatives to address antimicrobial resistance across various settings.

Transparency declarations

All the authors report no potential conflicts of interest.

Author contributions

YS and YT designed the study protocol. YS collected clinical data, and HH analysed the data and composed the figures. YS and YT drafted the first version of the manuscript. YT revised the manuscript, and all the authors contributed to the final version of the manuscript for submission.

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Declarations of interest

The authors declare no conflicts of interest and have submitted the ICMJE form for the Disclosures of Potential Conflicts of Interest.

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